

The Effects of a Large-Scale Mental Health Reform: Evidence from Brazil[†]

By MATEUS DIAS AND LUIZ FELIPE FONTES*

This paper studies the Brazilian Psychiatric Reform, which reorganized the public mental health care provision by introducing mental health centers (CAPS) as a community-based substitute for inpatient care. Our research design exploits the rollout of CAPS in a difference-in-differences framework. We show that these centers increased outpatient mental health care production and reduced psychiatric hospitalizations. These reductions were more pronounced for long-stay admissions and among patients with schizophrenia. We find that the savings implied by fewer admissions do not offset the cost of the policy. Also, the reform did not reduce mental health mortality and it increased violent crimes. (JEL H51, I12, I18, O15)

Mental disorders are a burdensome condition, having affected more than 1 billion people globally and being responsible for 19 percent of all years lived with disability in 2016 (Rehm and Shield 2019). Moreover, people with major depression and schizophrenia have a 40 percent to 60 percent greater chance of dying prematurely than the general population, and suicide is the second-most-common cause of death among young people globally (World Health Organization 2013). Mental health issues can also have severe negative economic consequences, disproportionately affecting the poor (Ridley et al. 2020). Despite such alarming patterns, health systems have not adequately responded to the burden of mental disorders; there is a significant gap between the need for treatment and its provision all over the world (World Health Organization 2022).

Despite broad consensus that access to mental health care is currently inadequate, debate over the best methods to provide such care continues. In the second half of the twentieth century, many countries transitioned from a model of mental health care centered on psychiatric hospitals to one based on community care. However, the

*Dias: Católica Lisbon School of Business and Economics (email: matdias@ucp.pt); Fontes: Insper (email: luiz.fontes@insper.edu.br). Matthew Notowidigdo was coeditor for this article. We thank the three anonymous referees for valuable feedback that allowed us to improve this article. We thank Bruno Ferman, Rudi Rocha, and Ilyana Kuziemko for their help and guidance at different stages of this project. We also benefited from comments and discussions with Rodrigo Soares, Nicolás Ajzenman, Daniel da Mata, Bladimir Carrillo, Antonio León, Janet Currie, Thomas Fujiwara, David Silver, Adriana Corredor-Waldron, Patrick Agte, Christiane Szerman, and seminar participants at the Essen Economics of Mental Health Workshop, iHEA Congress, Meeting of the Brazilian Econometric Society, FGV EESP, UFPE PIMES, and Princeton's CHW/RPDS student lunch. We would like to thank the Institute for Health Policy Studies (IEPS) for their help and hospitality. Fontes gratefully acknowledges financial support from São Paulo Research Foundation (FAPESP) Grant 2019/25473-5.

[†]Go to <https://doi.org/10.1257/pol.20220246> to visit the article page for additional materials and author disclosure statement(s) or to comment in the online discussion forum.

effects of this change are far from clear. Research shows that nonhospital treatments can be effective for some mental disorders at a low cost (Patel et al. 2007, 2011). However, the international experience reveals that reduced inpatient treatment combined with inadequate community-based care, without proper integration with other medical services, may generate undesirable consequences (Lamb 2015; Sisti, Segal, and Emanuel 2015; Rosenbaum 2016).¹

We investigate this question by examining Brazil's 2002 psychiatric reform. The Brazilian reform was centered on the introduction of Psychosocial Care Centers (*Centros de Atenção Psicossocial*—henceforth, CAPS) as a community-based substitute for inpatient care for people with mental disorders. More specifically, CAPS provide several psychosocial and outpatient procedures, such as talk therapies, occupational therapies, and medical consultations, and work as gateways to the public mental health care system.

We assess the effects of introducing a CAPS in a municipality on the density of mental health professionals and psychiatric beds, outpatient mental health care production, public spending on mental health, and hospital admissions and deaths by cause. As to the causes of deaths and admissions, we investigate mental and behavior disorders, suicide, illnesses caused by alcohol abuse, and overdose. Finally, we look at the effect on homicides, following the criminology literature that reports a correlation between deinstitutionalization and community-based care with violent crimes (Lamb 2015). Data limitations prevent us from evaluating the policy effects on well-being, which would require us to estimate the impact of the policy on less severe aspects of mental health.

Our empirical strategy exploits the rollout of CAPS across Brazilian municipalities in a difference-in-differences (DID) framework. Following De Chaisemartin and D'Haultfoeuille (2020), we use a DID estimator that is robust to heterogeneous treatment effects across cohorts and over time. Parallel pre-trends for the outcomes we evaluate support the design's internal validity. We first document that the introduction of CAPS increased access to and utilization of community-based mental health care. Specifically, implementing these centers was associated with immediate and significant increases in the density of mental health professionals and mental health outpatient visits. Consistent with these results, we also found an increase in drugs dispensed in outpatient care to treat psychiatric disorders.

Turning to hospitalization outcomes, we find that CAPS decreased hospital admissions due to mental illnesses. This reduction is driven by the reduction of long-stay hospitalizations, largely among individuals with schizophrenia, for whom CAPS may not be well equipped to provide care. As for public spending on mental health, we find a decrease in expenses related to mental health hospitalizations. However, based on the government expenditures on the program, our estimates indicate that the reform resulted in a net financial loss. Additionally, we find no effect of CAPS on deaths related to mental illnesses. Finally, we see a persistent increase in homicide rates following the reform. Although not definitive, these results align

¹ This idea is often connected with the concept of transinstitutionalization: when individuals are released from psychiatric institutions and no adequate alternative is offered, they may migrate to other institutions, usually part of the correctional system.

with the Penrose hypothesis, which states that inpatient care may impact violent crime through an incapacitation effect on patients more prone to violence, as either victims or perpetrators.

This paper contributes to the growing economic literature on mental health. Many authors study determinants of mental health, such as medication (Dalsgaard, Nielsen, and Simonsen 2014; Ludwig, Marcotte, and Norberg 2009), early life conditions (Persson and Rossin-Slater 2016; Almond and Mazumder 2011; Adhvaryu, Fenske, and Nyshadham 2019), economic shocks (Ruhm 2000; Schwandt 2018), and income shocks (Christian, Hensel, and Roth 2019; Baird, De Hoop, and Özler 2013). A recent set of experimental papers has evaluated the effects of psychological interventions on mental health and related outcomes (Baranov et al. 2020; Baranov, Haushofer, and Jang 2020). However, evidence of government policies' impact on mental health is scarce.

One of the few assessments of a government policy specifically aimed at mental health is that of Serena (2021), who examines the causal effects of increased insurance coverage for psychotherapy in Denmark. In contrast, we study the impact of increasing the supply of community-based alternatives to psychiatric institutionalization, a hotly debated policy choice that lacks solid empirical evidence. Although small-scale trials have studied the effectiveness of mental health care delivered in a community setting (Wiley-Exley 2007; Patel et al. 2007), the literature about the broader impacts of mental health policy changes at scale is still limited. This paper takes the first step in this regard. Importantly, we study this type of policy in a developing country, a setting in which the delivery of mental health care tends to be significantly hindered by a lack of resources.

This paper also contributes to the literature linking crime to mental health and services associated with it. Experts in criminology find that population-level crime rates correlate to mental health hospitalizations (Penrose 1939; Markovitz 2006; Raphael and Stoll 2013; Mundt et al. 2015). Similarly, medical research using individual-level data from discharged patients consistently reports a high prevalence of violent incidents involving former inpatients (e.g., Link, Andrews, and Cullen 1992; Walsh et al. 2003; Fleischman et al. 2014). However, most of these papers have focused on cross-sectional comparisons, which might be subject to omitted variable bias due to confounders affecting crime and inpatient care utilization.

The only attempt at a causal interpretation of the relation between mental hospitalization and crime is that of Landerso and Fallensen (2020). However, instead of studying hospital discharges, they analyze admissions at psychiatric hospitals and find that inpatient admittance reduces criminal behavior through incapacitation. Our paper exploits potentially exogenous changes in hospital admissions due to severe mental health conditions induced by the introduction of CAPS to study the causal effects of deinstitutionalization on violent crimes when alternative, community-based treatment becomes available.

The remainder of the paper is organized as follows. Section I summarizes the institutional and conceptual background. Section II introduces the data sources. Section III describes the empirical approach. In Section IV, we present and discuss our main results. Section V covers robustness checks. Section VI concludes.

I. Background

A. Institutional Background

Since the mid-twentieth century, many countries have shifted mental health care delivery away from psychiatric hospitals and toward community-based care, in what is known as deinstitutionalization. Proponents argue that community-based care is more humane and cost-effective (Rochefort 1984; Lamb and Bachrach 2001). This new paradigm influenced Brazilian psychiatrists, leading to the introduction of a bill to Congress in 1989 proposing the progressive substitution of psychiatric hospitals with community-based resources. The bill eventually became law in April 2001, but it was only made fully effective when the proposed CAPS were opened in 2002. According to the Brazilian Ministry of Health (Ministério da Saúde do Brasil 2015), this policy cost about 1.3 billion 2019 Brazilian Reais (BRL) between 2002 and 2014.² The total cost of the policy is comparable to the amount approved for mental health in 2019: 1.6 billion BRL. This amount is less than 1.5 percent of all approved health spending, and significantly less than the recent world average of 2.8 percent of health care spending (Biderman 2019).

The centerpieces of the psychiatric reform law are the CAPS, which are meant to be the primary source of care for the mentally ill. CAPS provide community-based substitute services for psychiatric inpatient care (Ministério da Saúde do Brasil 2005).³ CAPS are also the main gateway to the public mental health system, triaging and referring patients to other services as needed. At a CAPS, a patient is treated by a multidisciplinary team composed of psychiatrists, psychologists, occupational therapists, and social workers. This team prescribes a set of specific treatments, which can include consultations with a psychologist, medications, occupational therapy, and talk therapy.

To receive a CAPS, a municipality must apply to the federal government. The federal government has no formally established criteria to prioritize which municipalities receive CAPS. A municipality typically applies for a certain type of CAPS according to population size thresholds. There are three standard types: CAPS I (15,000–75,000 inhabitants), II (75,000–150,000 inhabitants), and III (over 150,000 inhabitants). In our data, these population criteria are important, but not strictly binding.

If a municipality already has a CAPS, it can also apply for one of two “special” types of CAPS: CAPS i, which focus on treating children and teenagers, and CAPS AD (I or III, depending on size), specialized in drug and alcohol abuse.⁴ When the application is approved,⁵ municipalities receive some financial support from the federal government: between 800,000 and 1,000,000 BRL for construction, and between

² Approximately US\$280 million.

³ CAPS existed even before the bill was presented. The first CAPS was created in the city of São Paulo in 1987. However, the number of CAPS before 2002 is negligible when compared to the number of centers created after the psychiatric reform law.

⁴ Having one of these specialized types nearby does not prevent standard CAPS from treating children and people suffering from drug or alcohol addiction.

⁵ In principle, applications can be rejected. However, we had access to the decisions made by the federal government in 2019, and very few proposals were rejected. The main reason for rejection was that the municipality did not meet the minimum population criterion established for the type of CAPS requested.